Journal of International Economics 87 (2012) 50–64

Contents lists available at SciVerse ScienceDirect

Journal of International Economics

journal homepage: www.elsevier.com/locate/jie

Getting beyond carry trade: What makes a safe haven currency?☆

Maurizio M. Habib, Livio Stracca *

European Central Bank, DG International and European Relations Germany

ARTICLE INFO

Article history:
Received 10 January 2011
Received in revised form 6 December 2011
Accepted 6 December 2011
Available online 20 December 2011

JEL classification:
E44
F31
G15

Keywords:
VIX
Global risk aversion
Safe haven currencies
Carry trade
Globalisation

ABSTRACT

The main objective of this paper is to investigate the fundamentals of safe haven currencies, which are those currencies that provide an hedge for a reference portfolio of risky assets, conditional on shocks to global risk aversion. We analyse a large panel of 52 currencies in advanced and emerging countries over almost 25 years of data. We find that only a few factors are robustly associated to a safe haven status, most notably the net foreign asset position, an indicator of external vulnerability, and whether currencies have been a good hedge in the past. In addition, the currencies of large, less financially open economies are a good hedge against global risk aversion shocks. By contrast, the level of the interest rate spread vs. the US is significant only during the latest crisis. Finally, we find some evidence of non-linearity as the importance of the fundamentals is stronger during crisis times.

1. Introduction

The global financial crisis of 2007–09 has renewed the public attention on safe haven currencies. As widely noted by observers, one paradoxical aspect of this crisis was the appreciation of the dollar as a safe haven currency exactly at the time in which the US was exporting a once-in-a-generation financial crisis to the rest of the world.

A relatively well-established literature has emphasised that returns on low interest rate currencies tend to be negatively correlated with global risk aversion, while high-yield currencies often crash exactly when global risk aversion is high (Brunnermeier et al., 2008). This leads to a systematic deviation from the Uncovered Interest Parity (UIP) whereby low interest rate currencies systematically under-perform except in exceptional circumstances, in particular when global exchange rate volatility is high (Menkhoff et al., 2012). However, this empirical regularity is not necessarily the same as safe haven status; the two concepts overlap only insofar as, and to the extent which, traders pursue carry trade strategies. In this paper, we want to go beyond this literature and try to establish what the "fundamentals" of safe haven currencies are. In short, what makes a safe haven currency?

Following Kaul and Sapp (2006), we define a safe haven currency as a currency which behaves like a hedge for a reference portfolio of risky assets conditional on movements in global risk aversion. If global risk aversion is high, the value of the reference portfolio of risky assets falls and a safe haven currency appreciates; the opposite happens when global risk aversion is low.

In particular, we put forward three possible sets of explanations of a safe haven status. First, a currency may be a safe haven if the country issuing it is itself safe and low risk. That may be appreciated by nervous investors in times of high risk aversion. Second, we surmise that size and liquidity of a country's financial market may support a safe haven status, an argument that has been called for during the latest financial crisis. When global risk aversion is high, market liquidity may dry up and most liquid markets may get an additional bonus. Third, we test whether financial openness and more generally financial globalisation is a determinant of a safe haven status. In addition to these fundamental determinants, we also test whether safe haven currencies may arise out of a self-fulfilling prophecy: currencies which have behaved as safe haven assets in the past appreciate in times of high global risk aversion, independent of other possible determinants.

Thus, the purpose of this paper is to answer the fundamental question of what makes a safe haven currency. We investigate the behaviour of a large sample of 52 currencies (51 bilateral exchange rates against the US dollar) between 1986 and 2009, focusing in particular

☆ We thank participants in a seminar at the ECB, Giancarlo Corsetti, Simone Manganelli, Harald Hau, Sergio Rebelo, Angelo Ranaldo and two anonymous referees for useful suggestions and comments. The views contained in the paper belong to the authors and are not necessarily shared by the ECB.

* Corresponding author.
E-mail address: livio.stracca@ecb.int (L. Stracca).

0022-1996/$ – see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.jinteco.2011.12.005

on the relationship between currency returns, economic and financial fundamentals and global volatility and risk aversion. An essential element of our analysis is to appraise whether and which of these possible determinants is a stable and robust predictor of safe haven behaviour.

Our study is related to two strands of literature. First, there is a fast growing body of literature on the relationship between the profitability of carry trade strategies and global risk aversion (Brunnermeier et al., 2008; Lustig et al., 2011; Menkhoff et al., 2012) as well as the relevance of fundamentals in devising carry trade strategies (Jorda and Taylor, 2009; Nozaki, 2010). Of particular interest for our paper is Menkhoff et al. (2012) who find that returns from carry trade strategies can be well explained by a single factor, namely global exchange rate volatility. [1] Higher yield currencies perform well when volatility is moderate or low, but can lead to potentially large losses in a few episodes of high to very high volatility. Brunnermeier et al. (2008) find that carry traders are subject to crash risk, i.e. the sudden unwinding of carry trades in periods when risk appetite and funding liquidity decrease. A consequence of this literature is that low yield currencies are generally safe havens in times of financial distress. While this result is important and relevant for our work, it leaves largely unexplained the question of *why* certain currencies emerge as safe havens in the first place, which is what we attempt to address here.

Second, there are other studies which have tried to detect safe haven currencies or their features, identifying ex ante the crisis event or the safe haven currency. For instance, Ranaldo and Soderlind (2009) find that the Japanese yen and the Swiss franc, but also the euro and British pound, may be regarded as safe havens during crisis episodes preceding the latest financial crisis. The 2008 financial crisis emerged as an important case study where safe haven effects went against typical patterns (Kohler, 2010) partially in contrast with the results of Ranaldo and Soderlind (2009). Indeed, specific factors may have been at play during the latest crisis. McCauley and McGuire (2009) stress the role of US dollar shortages — which were generated by the funding of net long US dollar exposure by European banks — and the role of "overhedged" US dollar positions — resulting from write-downs of US dollar assets — in supporting the US dollar exchange rate during the latest crisis. Indeed, Hui et al. (2010) confirm econometrically the impact of market-wide liquidity risk on exchange rate movements. Fratzscher (2009) finds that countries with low foreign exchange reserves, weak current account positions and high direct financial exposure vis-à-vis the United States depreciated the most during the crisis. However, Fratzscher's paper is limited to the 2007–09 financial crisis and does not offer a longer historical perspective.

This paper has the main purpose to merge these two strands of literature in a single empirical framework. In particular, the main contribution of the paper to the existing literature is to study thoroughly the relationship between currency movements and global risk aversion expanding (i) the sample of currencies, including up to 52 developed (23) and emerging (29) economies, (ii) the time period going as far as back 1986 and including the latest crisis and (iii) the set of potentially relevant explanatory and control variables, including policy, economic, financial and institutional factors. We also conduct an extensive analysis of stability and robustness, in particular in order to test whether, as commonly argued, the global credit crisis of 2007–09 has indeed different characteristics, in terms of safe haven currencies, compared with previous high global volatility episodes.

Generally speaking, we find few variables to be consistently and robustly significant in predicting a safe haven behaviour. That is not surprising given the large literature on the exchange rate disconnect, dating back to Meese and Rogoff (1983), which is the source of a still lively debate. [2] However, we do find that the interest rate spread is associated with a safe haven status only during the latest crisis; a result consistent with Kohler (2010). This confirms the notion that the interest rate differential is not a fundamental driver of safe haven status, and it depends on carry trade strategies being pursued. Self-fulfilling prophecies, instead, are found to be more important than carry trade strategies in explaining currency excess returns in times of financial stress. Turning to the economic fundamentals, the Net Financial Asset (NFA) position is by far the most consistent indicator that tends to be associated to safe haven behaviour, and this is the core result of our study. Moreover, the larger the size of the economy, or the stock market capitalisation, relative to world GDP, the higher the currency excess returns in times of financial stress. In contrast, the currencies of countries with more developed financial markets — relative to domestic GDP — or more financially open tend to be a bad hedge against global risk aversions shocks. Finally, we find some evidence of non-linearity, as the impact of the fundamentals that we identify is stronger in crisis than in normal times, as well as following rises (rather than declines) in global risk aversion.

The structure of the paper is as follows. In the next section we introduce the database. We present the empirical model is Section 3. Section 4 presents the results of the baseline model. Section 5 reports the results of an extended model, including a set of potentially relevant economic, financial and institutional variables, and further robustness checks. Section 6 considers the performance of different currency portfolios in crisis times, and, finally, Section 7 concludes.

2. Data

A large dataset of currencies, financial, economic and institutional variables has been created to study the behaviour of exchange rates during period of low and high global financial volatility. The dataset includes 51 monthly (period average) bilateral exchange rates against the US dollar from January 1986 until December 2009, which have been obtained from the IMF International Financial Statistics. Table 1 reports the list of countries used in the paper, sub-divided into advanced and emerging countries (respectively 23 and 29 of them); note that no observations are included for euro area countries since they join the euro area, which — while unavoidable in the context of our analysis — makes our panel strongly unbalanced. On the other hand, we introduce the euro as a new currency from 1999 onwards.

2.1. Our left-hand side variable

We compute monthly currency excess returns as follows,

$$\Delta e_{it} = log(e_{it}) - log\left(e_{i,t-1}\right) + \left(1 + R^{i}_{t-1}\right)^{\frac{1}{12}} - \left(1 + R^{USD}_{t-1}\right)^{\frac{1}{12}} \qquad (1)$$

where e_{it} is the bilateral exchange rate vs. the dollar, defined as dollars per unit of the given currency i, R^i_t is the annualised 1-month interbank interest rate for currency i, and R^{USD}_t is the same concept for the US dollar.

2.2. Global risk aversion shocks

The VIX index of the Chicago Board Options Exchange (CBOE) measuring the implied volatility of S&P 500 index options is our baseline measure of global risk aversion, as has become rather common in the literature. Indeed, previous papers have found that the VIX is highly correlated to many manifestations of risk and risk aversion (Collin-Dufresne et al., 2001). Starting from the VIX, we identify a measure of "global risk aversion shocks", denoted as v_t throughout the paper, which corresponds to (i) an *increase* in the VIX and (ii) a *fall* in the return of a global unhedged stock market portfolio (Campbell et al., 2010). The shock is identified by applying sign

[1] See also Christiansen et al. (2011).

[2] See Cheung et al. (2005) for a recent re-assessment of the empirical analysis of Meese and Rogoff (1983), basically confirming original results. See instead Engel and West (2005) for a reconciliation of the near-random walk behaviour of the exchange rate with fundamentals in a rational expectations present-value model.

restrictions to a bi-variate Vector Autoregressive (VAR) model including the two variables following the approach of Rubio-Ramirez et al. (2010).[3] By definition, therefore, if $v_t > 0$ we experience a situation in which (i) global risk aversion goes up and (ii) the return on a global unhedged stock market portfolio is negative. A currency appreciating in such a situation is defined as a safe haven currency. Table 2 reports the correlations between the global risk aversion shocks, the VIX and the returns of the global unhedged stock market portfolio.

The set of control variables includes a large number of policy, economic, financial and institutional factors which may be particular relevant in time of financial stress. These control variables have been divided in four main groups: (i) baseline variables which include the interest rate spread (emphasised in the carry trade literature) and a number of variables controlling for policy interventions; (ii) country risk variables; (iii) measures of size of the economy and size and liquidity of financial markets; (iv) measures of financial openness.

2.3. Baseline variables

The first group contains the interest rate differential between one-month domestic interbank and US rates, which captures the carry trade effects which have been emphasised in previous literature. We then have a set of control variables. One element that we need to take into account in our analysis is the degree of flexibility of the currencies, in particular whether currencies are pegged to the dollar, or any other currency (mainly the Deutsche Mark, DEM, before 1999 and the euro after 1999). Attempts by the authorities to keep the exchange rate stable vis-a-vis fluctuations in global volatility can drive a wedge between the fundamentals and the observed exchange rate behaviour, which could distort our estimates. We need, therefore, some control for exchange rate manipulation. One variable that we will use in the empirical analysis is an estimate of whether a certain currency i is de facto pegged to the dollar (or the DEM/euro) at time t. Similar to Levy-Yeyati and Sturzenegger (2005), we start from a de facto measure of exchange rate flexibility vis-à-vis the USD (or DEM/euro) defined as

$$FLEX_{it} = \frac{1}{12}\sum_{j=1}^{12} Abs\left(\Delta e'_{i,t-j}\right) \tag{2}$$

where $\Delta e'_{i,t}$ is the bilateral appreciation vis-a-vis the US dollar (or DEM/euro), not correcting for the interest rate differential. This cumulative absolute depreciation or appreciation in the previous year should give an idea of the degree to which a given currency is floating against the US dollar (or the DEM/euro).[4] Based on the $FLEX_{it}$ variable, we then construct a dummy variable PEG_{it} which takes value 1 if $FLEX_{it} < 1\%$ and zero otherwise. The threshold value of 1% is chosen based on the statistical distribution of the *FLEX* variable across time and countries, but we have experimented with other values, with similar results.

In addition, we include variables which explicitly capture the possible government measures aiming at offsetting the impact of higher (or lower) global risk aversion on their exchange rates. These are the growth in foreign exchange reserves *in the same month*, under the presumption that reserves are used to stem an appreciation or depreciation of a given currency vs. the US dollar, and the monthly change in the 1-month interbank interest rate, again in the same month. Manipulating short-term interest rates has been a way that several countries have used in the attempt to stem currency crises (for example during the European Monetary System and Asian crises). Note that we have included the former variable, the growth rate of foreign exchange reserves, both alone and in interaction with the *PEG* variable, the latter interaction based on the idea that variations in official reserves signal a desire to stem the exchange rate appreciation or depreciation only in pegging countries and not in others.

2.4. "Self-fulfilling" safe haven currencies

We consider the possibility that a safe haven currency arises out of a self-fulfilling prophecy. Currencies which have behaved as a safe haven in the past may be considered as safe haven at time t, independent of any fundamental determinant of safe haven status. To investigate this possibility empirically, we recursively compute the following variable,

$$z_{it} = Corr_{t_0,t-1}(\Delta e_{it}, v_t) \tag{3}$$

This correlation is computed between the beginning of the sample (t_0) and time $t-1$; a positive value for this variable identifies whether a certain currency has behaved as a safe haven on average in the past.

2.5. Country risk and vulnerability

The second group consists of those variables that are monitored by analysts when assessing country risk vulnerability, such as the inflation rate, the ratio of public debt to GDP, the current account of the balance of payments over GDP and net foreign asset position to GDP, the foreign exchange reserves to import ratio, and an indicator of banking crises from Laeven and Valencia (2008). In addition, we consider indicators of overall institutional quality of the country, namely the Rule of Law indicator from the World Bank Governance Indicators (available from 1996) and the ICRG Country Risk rating.

2.6. Size and liquidity of financial markets

A third group of variables includes all the controls for the size of the economy and size and liquidity of financial markets: the GDP weight in the world economy, private credit to the country's and world GDP, stock market capitalisation to the country's and world GDP and the bid–ask spread in the foreign exchange market, an indicator of liquidity of this market. By including these variables, we are able to control whether financial investors flee towards larger and more liquid markets in the midst of crises. At the same time, it may be that currencies with small financial markets which are usually considered as a safe haven — the Swiss franc, for instance — may be subject to larger price effects since strong demand is rationed by short supply of financial instruments denominated in that currency. The expected sign of these size/liquidity variables is therefore ambiguous.

2.7. Financial openness

Finally, a fourth group of variables consists of various measures of financial and foreign exchange openness. These include measures of *de facto* financial openness (such as the ratio of external financial assets and liabilities to GDP and foreign loans to GDP) as well as some *de iure* variables, namely the capital account restrictions as estimated in Schindler (2009). The latter are however available only for a relatively short sample period (1995–2005) and will therefore only be used as a robustness check.

The data have been obtained from a number of different sources: International Monetary Fund, World Bank, Haver, Datastream/Thomson Reuters and Barclays BBI, and International Country Risk Guide. Table 3

[3] The lag order in the VAR is imposed based on the Schwarz information criterion; the shocks computed from a VAR with a higher lag order were very similar. The identifying restrictions which are imposed to the bivariate VAR are those resulting from the identification scheme that is closer to the median of all impulse responses satisfying the restriction that the global risk aversion shock leads to a rise, on impact, in the VIX and a fall in the global unhedged stock market portfolio.

[4] In this respect, we believe that a de facto measure is more reliable than a *de iure* one, as many emerging countries have "fear of floating".

provides a complete and detailed description of all variables and the respective source. A number of control variables are only available at an annual frequency and have been interpolated to a monthly frequency through linear interpolation.

Table 4 reports descriptive statistics for all the variables used in the paper, for the full sample as well as for advanced and emerging economies separately. The number of observations varies considerably depending on the variable; for example, we only have about 5600 monthly data for capital account restrictions, but over 14,000 for inflation. This highlights the caveat that some of our regressions may be difficult to compare since they may refer to substantially different sample periods and country coverage. This limitation should be kept in mind in interpreting our results, in particular for the robustness analysis.

3. Empirical model

The main purpose of the paper is to study the safe haven status of currencies. As mentioned in the Introduction, we consider safe haven currencies those currencies which are a hedge for a (representative) market portfolio of risky assets conditional on shocks to global risk aversion, v_t. The general expression for our empirical model is

$$\Delta e_{it} = \alpha v_t + \beta x_{it} * v_t + \gamma x_{it} + \delta_i + \rho \Delta e_{i,t-1} + \varepsilon_{it} \tag{4}$$

where x is a vector with domestic variables that may affect the elasticity of returns to changes in shocks to global risk aversion (most of them actually timed $t-1$ to avoid simultaneity problems) and ε is a disturbance term. The main parameters of interest in our paper are contained in β, as we want to analyse the determinants of currency returns' reaction to changes in shocks to global risk aversion. Suppose, for example, that we want to test whether currencies with high (low) nominal short-term interest rates, R_{it}, deliver lower (higher) excess returns in times of high risk aversion. If the interaction term, $R_{it} * v_t$, is negative and significant, then we can conclude that the level of the nominal short-term interest rate influences the behaviour of exchange rates in relation to global risk aversion shocks. The negative sign can be interpreted as the reversal of carry trade strategies. Moreover, we consider both high frequency and low frequency, structural variables in the vector x. Note also that, differently from other papers based on case studies of crises, such as Fratzscher (2009), our analysis uses the full sample information to identify the impact of greater volatility and rising risk aversion on exchange rates. Finally, note that the v measure has been standardized (to zero mean and unit standard deviation) in order to facilitate the interpretation of the interaction terms, which may be thought of as marginal effects.

One potential concern that may arise in the estimation of Eq. (4) is reverse causality. For this reason, we generally include variables dated $t-1$, with the exception of the change in the foreign exchange reserves and of the change in the short-term interest rate, which are dated t due to their different role in the equation. For all variables that have been obtained from interpolation from annual data (say, public debt to GDP) we use the $t-12$ lag, in order to rule out the risk of reverse causality if data from the same year are used. Also note that the date t variables are included as controls and we make no statement about the causal interpretation of the estimated coefficient, for which an instrumental variable estimation would be needed.[5]

We will also be looking at specifications where we include extreme values and potential asymmetries of the impact of global volatility,

$$\Delta e_{it} = \alpha' \tilde{v}_t + \beta' x_{it} * \tilde{v}_t + \gamma x_{it} + \delta_i + \rho \Delta e_{i,t-1} + \varepsilon_t \tag{5}$$

where $\tilde{v}$ is a truncated function of global risk aversion, e.g. very large shocks to global risk aversion or shocks happening at a particularly high or low starting value for the VIX (say, a shock happening in a crisis may have different effects from the same shock in a tranquil period).

It is self-evident that the contemporaneous inclusion of all control variables would lead to an overparametrised model. For this reason, we adopted an incremental approach to the estimation of Eq. (4). First, in the baseline model, Eq. (4) is estimated in a parsimonious specification, including only the control variables in the first baseline group, as well as country fixed effects and the own lag of the currency excess returns:

$$\begin{aligned}\Delta e_{it} = {} & \alpha v_t + \beta\left(R_{i,t-1} - R^{US}_{i,t-1}\right) v_t + \beta' x_{it} * v_t + \gamma\left(R_{i,t-1} - R^{US}_{i,t-1}\right) \\ & + \gamma' x_{it} + \delta_i + \rho \Delta e_{i,t-1} + \varepsilon_t\end{aligned} \tag{6}$$

where $x_{it} = [\Delta reserves_{it}, \Delta(R_{it} - R^{US}_{it}), PEG_{it}]$ is the vector of controls, R represents the one-month rate, and $\Delta reserves_{it}$ is the growth rate of FX reserves, and PEG is a dummy variable as defined above, taken separately for the US dollar, the euro and the DEM. In a variant of the model, we also add $\Delta reserves_{it} * PEG_{it}$ to the x vector.

According to the UIP, countries with a higher nominal interest rate than the US at time $t-1$ should tend to depreciate against the US dollar over time. However, many studies found that this coefficient has the "wrong" positive sign, rejecting the validity of the UIP, in particular for advanced economies with floating exchange rates and over shorter horizons.[6] Global risk aversion shocks drive a wedge in the UIP, which is captured by the parameter β. If β is negative and significant, countries having a higher nominal interest rate than the US deliver lower excess returns when v is high, i.e. in times of high global financial market volatility and negative global stock market returns.

Is it sufficient to have a low nominal interest rate to be a safe haven currency? Of course, the nominal interest rate may be high or low depending on domestic macroeconomic conditions and the monetary policy regime prevailing in individual countries. For example, a country with low inflation and economic growth will tend to have a low nominal interest rate. In an open economy, however, the nominal interest rate is also an endogenous variable which responds to prevailing conditions in global financial markets. Suppose that a certain country, say Switzerland, is preferred by investors in times of global financial stress due to its intrinsic, fundamental characteristics (say, political stability). In this case, there will be an inflow of capital in the country which would tend to make its currency appreciate, even if the domestic interest rate is low. In this case, it is not the level of the interest rate in itself that attracts investors, but rather the underlying fundamental that allows the country in question to enjoy a combination of exchange rate appreciation and low interest rate at times of heightened global risk aversion. In other words, the level of the nominal interest rate might be a summary indicator of a number of unobservable country characteristics, which we want to tease out explicitly in this paper.

Once we obtain a parsimonious specification of the baseline model in (4), we use this to test the statistical significance of the other groups of control variables (gradually expanding the x_{it} vector), taken one at the time. In each step, the new group of control variables is first included in the baseline model — each variable separately and then jointly, whenever they are significant individually — without variables from other groups. In the following step, the model is extended with a new group of variables, first included one by one and then jointly, including variables of other groups that are statistically significant in the previous stage. Once we have a final model possibly including a small set of variables from all groups, we tinker with the

[5] We carried out some estimations using instrumental variables (GMM) — not reported for brevity — and these led to similar results as the pooled OLS, but with somewhat less precision in the estimates.

[6] See Chinn (2006) for a recent critical review of the literature on the UIP puzzle.

model by analysing its robustness to different samples as well as possible non-linear versions of the model.

The models in Eqs. (4)–(6) are estimated, throughout the paper, by OLS panel fixed effects, using Driscoll and Kraay (1998) standard errors in order to account for temporal and cross-sectional dependence. The choice of fixed effects is a natural one in a country panel where it may be difficult to assume that the individual constant terms are randomly distributed. Empirically, robustness checks show that there are no major differences in the results between pooled and fixed effects estimations. Therefore, a model estimated with pooled OLS would have been equally suitable. Eventually, we decided to report the panel fixed effects results since the Wald test rejects the hypothesis that all country fixed effects are jointly equal to zero. The time dimension of our panel is very large, up to 288 monthly observations in the best case, and would therefore be able to deal with the potential bias induced by panel estimations in a dynamic setting, which is of the order $1/T$ (see Nickell, 1981).

4. Baseline results

The first step of our empirical investigation is to specify a parsimonious baseline model including the reversal of carry trade strategies and the role of self-fulfilling prophecies based solely on the behaviour of currency returns in previous crises, as well as potential policy measures adopted by countries which peg or fear of floating vis-à-vis the US dollar. Table 5 reports the results for the estimation of this baseline model allowing for a different impact across advanced or emerging economies (columns 3 and 4) and studying how the relationship could have changed in the latest crisis, from August 2007 onwards (column 5 versus 6), or after the introduction of the euro since 1999 (column 7 versus 8). It is important to consider all these possible variants in order to have an appreciation of the robustness of the main results, which is a main objective of this paper.

The autocorrelation coefficient of the exchange rate is positive and significant, but this is mainly a mechanic consequence of taking monthly average data. The model generally explains up to about 15% of the variability within each individual country.[7]

The coefficient of the interaction of the VIX index with the interest rate spread versus the US dollar is negative and significant only in the sub-sample starting from August 2007 that includes the latest crisis. Moreover, even though not statistically significant, the coefficient is negative for advanced economies and positive for emerging markets. In other words, we find some evidence of a reversal of carry trade when global volatility is high only in the latest crisis and, possibly, across advanced economies, not for emerging markets. Indeed, only in the most recent years, currencies of emerging markets such as the Brazilian real or the South African rand have been included in carry trade strategies, whereas many other emerging market currencies may not have sufficient liquidity to support carry trade speculation. Our results are consistent with the findings of Kohler (2010), who shows that interest rate differentials explain more of the crisis-related exchange rate changes in 2008–09 than in the past, and Bansal and Dahlquist (2000) view that the often found negative correlation between the expected currency depreciation and interest rate differential is confined to developed economies.

Interestingly, self-fulfilling prophecies appear to play an important role in explaining currency returns during turbulent periods. The coefficient associated with our proxy for this particular determinant — the recursive correlation between currency excess returns and global risk aversion shocks up to time $t-1$ — is positive and statistically significant when interacted with the global risk aversion shock at time t. Currencies whose excess returns have been positively correlated with global risk aversion shocks up to $t-1$ i.e. that have been in the past a good hedge to declining global stock markets and rising financial volatility — deliver higher currency excess returns when these global risk aversion shocks are positive also at time t.

Our results also confirm that it is important to control for the policy measures associated with exchange rate pressure: changes in foreign exchange reserves; changes in the interest rate spread; and the de facto flexibility of the exchange rate. The first one, changes in foreign exchange reserves, does not have a statistically significant impact when interacted with global risk aversion shocks. However, the other two control policy variables are statistically significant and with the expected sign. An increase in the interest rate spread versus the US interest rate at time t is associated with higher currency excess returns in turbulent periods. In any case, we emphasise again the fact that the coefficients associated to these two variables cannot be interpreted in a structural manner, due to the possibility of reverse causality. Similarly, we find that the *PEG* variables are positive and often statistically significant in particular for the US dollar and the Deutsche mark, not for the euro. Unsurprisingly, the sign is positive, suggesting systematic appreciation of those currencies pegging to main international currency anchors compared with the floating ones. Note that these variables are mostly insignificant when included on their own, i.e. when the standardized global risk aversion shock is at its sample mean of zero.

Overall, based on these results, the revised baseline model will, from now on, include the interest rate spread, which is barely significant but important as theoretical benchmark, its contemporaneous change and the *PEG* variables for the US dollar and the Deutsche mark.

Before concluding this preliminary analysis, it is interesting to look at the coefficient associated with the global risk aversion shock itself. In the way our empirical model is constructed, this variable essentially captures the performance of an equally-weighted portfolio of foreign currencies against the US dollar depending on the level of this global risk aversion shock. A negative coefficient indicates a decline in the excess return from this foreign currency portfolio, corresponding to a dollar *appreciation* above any potential change in the interest rate differential in favour of the foreign currencies. In other words, the coefficient on v_t captures the specific behaviour of the US dollar as such. Interestingly, this coefficient is not statistically significant *before* the 2007–09 global financial crisis, in line with the results of other studies such as Diekmann and Meurers (2007), but negative and significant *during* the last crisis, confirming the perception that the recent behaviour of the US dollar has been rather anomalous compared with previous regularities.[8] This result is robust also when conditioning to other variables, as we will see later on. We therefore conclude that, contrary to the common belief, which has been strengthened by recent events, the US dollar is not always a safe haven currency.

5. Identifying the fundamental features of safe haven currencies

After having specified a parsimonious model of currency returns and global risk aversion, we are now in the position to test the statistical significance of a large set of economic, financial and institutional variables potentially affecting currency returns, in particular when global financial volatility changes.

5.1. Controlling for country risk and vulnerability

As mentioned in the Introduction, one key ingredient of a safe haven currency may be that the country issuing the currency and therefore the bulk of the financial instruments denominated in that

[7] Fixed-effects regressions with the Driscoll–Kraay standard errors estimate the pooled OLS of the within-transformation of the model variables. For this reason, tables report only the *within* R2. Fixed-effects regression excluding the Driscoll–Kraay correction show that the between variability contributes only marginally to the overall variability of the panel, which is dominated by the within variability.

[8] Note also that, when volatility is high, the currencies pegging to the US dollar tend to *depreciate* in the sub-sample until 1999.

currency is seen as “low risk” by nervous investors. Hence, Table 6 presents the baseline model extended with a number of variables measuring country risk and vulnerability. These control variables are first entered individually in separate regressions; in the last columns of the table, we then present one possible specification including jointly the variables that proved to be statistically significant when entered individually. The results show that indicators of external sustainability such as the net foreign asset position or the current account are statistically significant when interacted with the global risk aversion shock. As expected, the sign is positive for both variables, indicating that countries with better external positions — irrespective of whether one takes a flow or stock perspective of external sustainability — tend to have currencies that appreciated with rising global risk aversion and falling global stock markets (safe havens). Similarly, the coefficient associated with the ICRG country risk rating, a composite indicator of political, economic and financial risk, is positive and statistically significant when interacted with the global risk aversion shock, indicating that the currencies of countries with better ratings tend to be a good hedge in turbulent periods. Other variables, such as the inflation rate, the public debt to GDP ratio, the ratio of foreign exchange reserves to imports, the Rule of Law indicator and the dummy variable capturing whether a country is experiencing a banking crisis are not significant. When put jointly, only the NFA position remains statistically significant, with the expected sign; the current account and the country risk rating are insignificant. As a result, in the following steps, the baseline model is augmented with the NFA position, the variable among the macro risk and vulnerability indicators that appears to be the most robust in explaining the safe haven status.

It is important to test for advanced and emerging countries separately as the determinants of safe haven (or un-safe haven) status may be different between them. These results are not shown for reasons of space, but it is interesting to highlight a few differences between advanced and emerging countries.[9] First, the coefficient associated with the public debt to GDP ratio is (unexpectedly) positive and significant for advanced economies, while (as expected) negative and significant for emerging markets. In a similar fashion, the country risk rating is (unexpectedly) negative, but not statistically significant, for advanced economies and positive (as expected) and statistically different from zero for emerging markets. The coefficient associated with the interest rate spread is in general not statistically significant, even though the sign is negative (carry trade reversal) and significant in one specification for advanced economies and positive, but never statistically different from zero, for emerging countries. Finally, the ratio of foreign exchange reserves to imports is significant for advanced countries with the expected sign, but (quite surprisingly) not in emerging countries.

5.2. Controlling for the size of the economy and the size and liquidity of financial and foreign exchange markets

In the next step, we further extend the baseline model with an additional set of variables controlling for the size of the economy and the size of liquidity of financial and foreign exchange markets — stock market capitalisation, private sector credit, both as a share of domestic and world GDP, and the bid–ask spread in the foreign exchange market. Which of these elements contributes to the making of a safe haven currency? Table 7 presents the results of these additional regressions. A number of indicators of the relative or absolute size of the economy and the financial markets turn out to be statistically significant. A clear trend appears from the sign of the coefficients of the variables interacted with the global risk aversion shock. Indicators of the relative size of the financial markets as a ratio to domestic GDP, such as stock market capitalisation or private sector credit, have a negative sign, suggesting that the currencies of countries where financial markets are domestically important tend to be associated with *negative* excess returns in turbulent periods. However, currencies of economies or financial markets that are large relative to world GDP deliver *positive* excess returns when global risk aversion rises. In other words, having a large stock market in relative domestic terms doesn't lead to safe haven status, but having a large economy or stock market in absolute terms does.

When put together, only the stock market capitalisation to domestic GDP (negative sign) and to world's GDP (positive sign) remain significant. The stock market capitalisation to world GDP is highly correlated with the relative size of the economy to world GDP, which explains why the latter variable is not statistically significant and changes sign in the joint specification. After a brief specification search, two possible models emerge as the most plausible: one including the size of the stock market (or the economy) relative to world GDP together with the stock market capitalisation to domestic GDP, and one more parsimonious including only the size of the economy (or the stock market) relative to world GDP. The time series for stock market capitalisation are shorter than for GDP, reducing the sample by more than 900 observations. For this reason, in the following step, we augment the baseline model only with the size of the economy relative to world GDP, even though the alternative model is equally valid and, if used, produces similar results.[10] In all possible specifications, the role of self-fulfilling prophecies and NFA positions turns out to be very important and statistically robust in order to explain the safe haven status of currencies. There are no major differences in the sign and statistical significance of the coefficients between advanced and emerging economies (not reported).

5.3. Controlling for financial openness

Finally, the last group of variables to be added to the baseline model consists of measures of financial openness such the sum of external assets and liabilities as a share of GDP, the foreign loans and international debt to GDP ratios from the World Bank database of Financial Development and Structure and a measure of *de jure* restrictions on cross-border financial transactions from Schindler (2009).

Table 8 shows that all de facto measures of financial openness are *negatively* associated with the safe haven status of currencies. This implies that the currencies of countries de facto open to capital flows tend to deliver negative excess returns in the wake of higher global risk aversion. In particular, the coefficients relative to financial openness, international debt and foreign loans are negative and statistically significant when interacted with the global risk aversion shock. When including financial openness and international debt jointly, the two variables with the longest available sample period, only the coefficient relative to financial openness, i.e. the sum of external assets and liabilities as a share of GDP, remains statistically significant. In addition, among the control variables previously included in the baseline model, the coefficient associated with the size of the economy appears to be less robust and not always statistically significant.

Variables measuring *de jure* capital restrictions as estimated by Schindler (2009) — where a higher score indicates a more restricted capital account — are generally not statistically significant. However, when running separate regressions for advanced and emerging economies (not shown), the coefficient associated with inflow capital restrictions in advanced economies is negative and statistically significant, indicating that these restrictions result, ceteris paribus,

[9] Results for advanced and emerging countries are available from the authors.

[10] Results, including all the subsequent robustness tests, for the alternative model including stock market capitalisation to world and domestic GDP are available from the authors. In this specification, the variables proxying for financial openness are not significant.

in negative currency returns, an outcome somewhat in contrast with that of the *de facto* measures of financial openness. For emerging markets, instead, the coefficients of the *de jure* measures of financial openness are positive, even though not statistically significant. However, we emphasise that these *de jure* indices are available for a very limited sample period and, hence, this result should be interpreted with caution.

Overall, we find that the 'best' model in order to explain the determinants of safe haven status includes (i) whether a currency has been in the past a good hedge, on average, against global risk aversion shocks (self-fulfilling prophecies); (ii) whether countries are pegged to an international currency (the US dollar or the euro/DEM); (iii) most notably, the NFA position, an indicator of countries' external vulnerability: the currencies of countries with a more positive net position perform clearly better in times of high global volatility; (iv) financial openness, measured by the sum of external assets and liabilities as a share of GDP: the lower financial openness, the better the safe haven properties of a given currency are.

Finally, it must be noted that after accounting for economic fundamentals, the level of short-term interest rate spreads vs. the US does not appear to be a significant determinant of the safe haven status, even though it is possible to detect a contemporaneous correlation between the *change* in interest rate spreads and the currency returns, which does not necessarily imply causation but which we interpret as mainly the reaction of the authorities to the threat of a sharp depreciation or appreciation in those countries which care about the stability of their exchange rate vs. the US dollar.

5.4. Robustness

In this section we test the robustness of our "final" specification by benchmarking our results against a number of different models (pooled estimates, a simple random walk, or excluding policy intervention variables), considering periods when the VIX increases dramatically and differentiating between negative, positive and very high levels of the global risk aversion shock. Finally, we test the robustness of our best model, varying the sample period and analysing the equation for advanced and emerging countries separately as already done for the baseline model in Table 5.

5.4.1. Benchmarking and extreme values of the VIX or global risk aversion

It is useful to try to quantify the effect of the control variables that have been identified by our analysis in order to understand the economic, rather than statistical significance, of our results for currency returns. One way to appreciate the economic significance of the proposed 'fundamental' determinants of safe haven status is to compare the R-squared of our best model in Table 9 (first column to the left) with that of a simple random walk model (second column), a random walk augmented with only interest rate differentials (carry trade model in the third column) or a model including our safe haven fundamentals (fourth column). It can be seen that the R-squared are of similar size, which indicates that interest rate differentials or the fundamental values explain relatively little of the overall variability of exchange rates. This is of course hardly surprising given previous results in the exchange rates literature; however, this is different in crisis periods, as we will see shortly. In addition, it is possible to note that the interest rate differential is statistically significant in interaction with global risk aversion shocks when there are no other control variables in the model, but differently from fundamental values that are robust to alternative specifications, its statistical significance disappears as other variables are added. This suggests that the interest rate spread with the USD may actually proxy for other variables that are captured in our full-blown model and it may be necessary to go beyond carry trade to understand the true nature of safe haven currencies.

So far, we have estimated a linear model in which the variables of interest have been interacted with the global risk aversion shock. In this sub-section, we investigate the possible role of extreme values of the VIX or the global risk aversion shock. In particular, similar to other event-based studies, we propose a definition of "distress period" based on the VIX. In our case, the distress episode begins in the month in which the percentage change in the VIX is above two standard deviations and ends when the level of the VIX falls below its long-term average plus one standard deviation. This definition identifies rather precisely the major crisis episodes, corresponding to more than 30 monthly observations in the VIX series.[11] The last five columns of Table 9 present estimates, again of the 'best' equation, outside and during all distress episodes according to the definition just proposed (column 5 and 6, respectively), differentiating between negative or positive global risk aversion shocks (columns 7 and 8, respectively) and global risk aversion shocks being in the highest decile (column 9), in order to investigate possible asymmetry and threshold effects. The estimates reveal (e.g. by comparing columns 5 and 6, or columns 7 and 8) that our model fits much better the sample including distress episodes or positive global risk aversion shocks rather than periods when there is no financial distress or negative risk aversion shocks. All in all, therefore, there is some evidence of nonlinearity in the data: the model is more informative for safe haven behaviour of currencies in periods of distress. This is also visible when comparing the R-squared of the full model with that of a naive benchmark (such as the random walk); in this case (not reported), the improvement in the R-squared is very large, indicating that the fundamentals that we identified are economically significant determinants of safe haven behaviour in crisis times.

5.4.2. Country group and sample periods

Table 10 reports on our best specification of the baseline model including (i) the NFA position and (ii) financial openness, measured as the sum of external assets and liabilities as a share of GDP. We first present the estimates for the full sample, and then move to split the country group into advanced and emerging, and then the sample before and after the August 2007 crisis and the introduction of the euro in 1999. Several interesting results emerge. First, we find again that the interest rate spread (as well as its change) only matters for the recent crisis, even though it is now also statistically significant at the 10% level in the period before the introduction of the euro. Second, in a similar fashion, the role of self-fulfilling prophecies seems to be more important in the later part of our sample. Third, the coefficient associated with the global risk aversion shock — which, as previously noted, identifies the behaviour of the US dollar specifically — shows again some instability over time. In particular, it is not statistically different from zero before 1999 or August 2007 (no safe haven status for the US dollar as such), but it is negative and significant since 1999 (US dollar is a safe haven). The result on the safe haven status of the US dollar is therefore not robust. Finally, we find that the NFA position is the most consistent fundamental determinant of the safe haven status, clearly outperforming all the other potential control variables in terms of robustness.

6. Comparing carry trade and NFA-based portfolios

As just noted, a main result of this paper is that the NFA position is a key fundamental driver of safe haven currencies. This begs the question of whether this variable can be used to operationalise a safe

[11] The distress episodes (start–end date in parenthesis) are the following: US stock market crash (10/1987-03/1988); 1st Gulf war (08/1990-10/1990); spillover of Asian crisis (11/1997-11/1997); Russian default (08/1998-10/1998); September 11th attack (09/2001-10/2001); internet bubble burst and 2nd Gulf war (07/2002-03/2003); US subprime mortgage crisis (03/2007); liquidity crunch (08/2007); and Lehman Brothers bankruptcy (09/2008-06/2009).

haven-currency portfolio, similar to the analysis performed by Menkhoff et al. (2011) on carry trade portfolios. In the same spirit of Menkhoff et al. (2011), we construct the returns on a synthetic portfolio of currencies which is, at time *t*, equally-weighted and *long* on currencies having an NFA position at least one standard deviation above the average across countries at time $t-12$, and *short* on currencies with an NFA position below at least one standard deviation of the same cross country average. We also construct the returns on a portfolio that is short on high (one-month) interest rate currencies (vs. the US interest rate) and long in low interest rate ones.[12] How do these portfolios perform in crisis times, i.e. as safe haven assets?

In Table 11, we report the coefficient on a regression of selected returns on the crisis dummy that we used in Table 9; the model also includes a dummy for the post-1999 period (since the exchange rates of the euro area countries collapse into the euro after 1999 and there is therefore a large change in the composition of the panel) as well one lag of the endogenous variable. It can be seen that, in crisis times, stock markets fall, the VIX rises sharply, and the Japanese yen and the Swiss franc act as safe havens vs. the US dollar (but not the euro). We find that both the NFA-based portfolio and the carry trade one are good hedges of stock market developments in crisis times. While the return on the carry trade portfolio is higher, it is also more volatile, and it may therefore be more convenient, depending on investor preferences, to build a safe haven portfolio of currencies based on the NFA position one year earlier.[13]

7. Conclusions

In this paper we have searched for the fundamentals of safe haven currencies, which are those currencies that behave like a hedge for a reference portfolio of risky assets conditional on movements in global risk aversion. Previous literature has uncovered the fact that systematic deviations from the Uncovered Interest Parity (UIP) may be attributed to a "crash risk" whereby some high-interest currencies depreciate sharply in times of financial stress, while low interest currencies typically appreciate (safe haven currencies). The first objective of this paper is to document this stylised fact on a large sample of 52 currencies (51 bilateral exchange rates), on a sample period spanning almost 25 years. Our main finding here is that there is evidence of carry trade reversal during the latest crisis, while other factors seem to be more important and robust in order to identify safe haven currencies across the whole sample.

As the next and more innovative step in our analysis, we look at the fundamental determinants of the individual currencies' loadings on the "crash risk" factor. What makes a currency a safe haven, hence with a lower return in good times and a higher return in periods of financial stress? We put forward three possible explanations. First, the loading may be related to the intrinsic risk profile of the country issuing the currency. A country that is intrinsically less risky may be preferred in times of higher global risk aversion. Second, it could be that currencies are safe havens if they are supported by a large country and by large, well developed and liquid financial (including foreign exchange) markets. Again, these characteristics may be desirable to investors in times of high risk aversion and low liquidity, while they may be not in more normal times. Finally, countries that are more open to the rest of the world, in particular in the financial market, may be differently affected by global turbulence. In particular, we surmise that more financially open countries may be more exposed to financial turbulence originating at a global level, and therefore be less likely to be safe havens. An ideal safe haven should be a place that is insulated from the global storm when the storm strikes; a difficult feat in times of financial globalisation.

We look at a large set of potential explanatory variables and countries but we find very few variables entering consistently and robustly as determinants of safe haven status. Moreover, even those which enter significantly have a rather small effect on monthly currency excess returns. This is a result which should suggest some caution against over-interpreting exchange rate movements in times of global stress, at least at a monthly frequency as in our analysis. Nonetheless, we do find a bunch of variables to be statistically significant and reasonably robust.

One main contribution of the paper to the existing literature is to put the spotlight on the fact that it is not the interest rate spread, as emphasised in the carry trade literature, the most consistent and robust predictor of safe haven status, but the NFA position, which is an indicator of country risk and external vulnerability. In addition, self-fulfilling prophecies play a more consistent role in explaining currency returns during turbulent periods than carry trade strategies. Currencies that have been in the past a good hedge to declining global stock markets and rising financial volatility are likely to remain a good hedge in the future, even after controlling for many other possible determinants of safe haven status. Notably, we find that the currencies of large economies, or with large financial markets, on a global scale, but less leveraged and less open to capital flows tend to have better hedging properties in time of financial distress. Moreover, it should be noted that the fundamentals of safe haven currencies may be different for advanced and emerging markets. For instance, the public debt to GDP is negatively associated with the safe haven status of the currencies of emerging markets, but the opposite holds true for advanced economies. Evidently, at least in the past crises (though certainly not more recently), investors scrutinised the sustainability of public finances of emerging markets much closer than that of advanced economies. Finally, we find some evidence of asymmetry and non-linearity in the model: the effects appear to be stronger in crisis periods and after positive, rather than negative, shocks to global risk aversion.

The analysis conducted in this paper is in-sample. A useful extension of our work would be whether it is possible to predict safe haven behaviour out of sample. Based on information up to time $t-1$, can a trader predict which currencies will appreciate if global volatility goes up by x%? Based on the results of our paper this appears to be a very difficult thing to do, but which could be very interesting to take up in future research.

Appendix

Table 1
List of economies.

Advanced (23)	*Emerging (29)*
United States; United Kingdom; Austria; Belgium; Denmark; France; Germany; Italy; Netherlands; Norway; Sweden; Switzerland; Canada; Japan; Finland; Greece; Iceland; Ireland; Portugal; Spain; Australia; New Zealand; Euro area	Turkey; South Africa; Argentina; Brazil; Mexico; Venezuela; Israel; Hong Kong; India; Indonesia; Korea; Malaysia; Philippines; Singapore; Thailand; Bulgaria; Russia; China; Czech Republic; Slovak Republic; Hungary; Croatia; Slovenia; Poland; Romania; Taiwan; Estonia; Latvia; Lithuania

Table 2
Correlation matrix of the Global Risk Aversion (GRA) shock, VIX and returns on global stock market index.

	GRA shock	*Return on global stock market index*	*VIX*
GRA shock	1		
Return on global stock market index	−0.69	1	
VIX	0.92	−0.60	1

Note: the table reports correlation coefficients over the sample 1986:1–2009:12. See Table 3 for definitions and sources.

[12] Note that unlike Menkhoff et al. (2012) we do not consider transaction costs in this calculation.

[13] In separate regressions, not reported for brevity, we also analysed the elasticity to GRA shocks. We also find that both the NFA and the carry-trade portfolios are safe havens (meaning that returns are positive when GRA shocks are positive), especially so in crisis times.

Table 3
Description of variables and sources.

Variable	Description	Source	Frequency
Exchange rate	Average bilateral nominal rate vs. USD. US dollars per unit of national currency	IMF International Financial Statistics (IFS)	M
Interest rate	1-month interbank intrest rate (or closer substitute)	Datastream	M
Currency excess returns	Exchange rate return minus 1-month interbank interest rate differential	Own calculations	M
Global unhedged stock market index	MSCI World stock market index of 1500 'world' stocks in 24 advanced economies. Data are in local currency.	Datastream	M
Risk aversion measures			
VIX	Implied volatility of S&P 500 index options (extend with VXO before 1990)	Haver/Chicago Board Options Exchange	M
GRA shock	Global Risk Aversion shock obtained as a shock which (i) leads to an increase in the VIX and (ii) leads to a fall in the return of a global unhedged stock market portfolio (see text)	Own calculations	M
Group 1: Speculative activity, exchange rate flexibility and policy measures			
Interest rate spread vs. USD	Spread between 1-month interbank rate (or closer substitute) and US 1-month interbank rate. Series extended using forward premium (see below)	Datastream	M
Forward premium	Difference between 1-month forward exchange rate and spot rate divided by spot rate, annualised	Datastream: Thomson Reuters and Barclays BBI	M
Growth of international reserves	Foreign exchange reserves. Log difference between time t and $t-1$	IMF IFS	M
Exchange rate flexibility vs. the USD	Computed as the average absolute depreciation or appreciation of a given currency vs. the USD in the preceding 12 months	Own calculations	M
Group 2: Country risk variables			
Inflation	Annual change in the consumer price index	IMF World Economic Outlook (WEO)	A
Public debt to GDP	Ratio of public debt to GDP	IMF WEO	A
Net foreign assets to GDP	Ratio of net foreign assets to GDP	IMF IFS and WEO	A
Current account to GDP	Ratio of current account to GDP	IMF WEO	A
FX Reserves to import	Ratio of foreign exchange reserves (M) to imports of goods and services (A)	IMF IFS and WEO	M/A
Country risk rating	Composite political (50%), economic (25%) and financial (25%) risk rating ranging between 0 (highest risk) and 100 (lowest risk)	International Country Risk Guide	A
Banking crisis	Dummy variable (1 = crisis, 0 = no crisis)	Laeven and Valencia (2008)	A
Rule of law	Includes several indicators which measure the extent to which agents have confidence in and abide by the rules of society	World Bank	A
Group 3: Size of the economy and size and liquidity of financial markets			
GDP Weight	PPP weight of world GDP	IMF WEO	A
Stock market capitalisation	From the World Bank database of Financial Development and Structure	World Bank	A
Private credit to GDP	From the World Bank database of Financial Development and Structure	World Bank	A
Stock market capital. to world GDP	From the World Bank database of Financial Development and Structure	World Bank	A
Private credit to world GDP	From the World Bank database of Financial Development and Structure	World Bank	A
Bid–ask spread	Spot exchange rate: ask-price minus bid-price divided by mid-price	Datastream: Thomson Reuters and Barclays BBI	M
Group 4: Financial openness			
Financial openness	Ratio between the sum of external financial assets and liabilities and nominal GDP in USD	IMF IFS and WEO	A
International debt to GDP	From the World Bank database of Financial Development and Structure	World Bank	A
Foreign loans to GDP	From the World Bank database of Financial Development and Structure	World Bank	A
Capital account restrictions	See Schindler (2009)	IMF	A

Table 4
Descriptive statistics.

	Full sample					Advanced economies					Emerging economies				
Variable	*Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*	*Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*	*Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*
MSCI LC returns (%)	288	0.4288	3.7535	−22.725	10.866	–	–	–	–	–	–	–	–	–	–
Global Risk Aversion (GRA) shock	286	0.0000	1.0000	−2.2487	5.7936	–	–	–	–	–	–	–	–	–	–
Currency excess return (%)	9944	0.2248	2.9604	−72.675	30.386	4700	0.2061	2.5602	−21.375	10.032	5244	0.2415	3.2780	−72.675	30.386
Interest rate spread vs. US (%)	11526	3.8459	9.6870	−6.744	199.411	6164	1.1377	3.3765	−6.559	29.895	5362	6.9591	13.0575	−6.744	199.411
Recursive correl. (GRA shock; Curr. returns)	13540	0.0666	0.1563	−1.0000	1.0000	6093	0.1381	0.1205	−1.0000	1.0000	7447	0.0081	0.1579	−1.0000	1.0000
Growth of international reserves	13344	0.0076	0.0263	−0.4333	0.3851	6180	0.0021	0.0209	−0.1852	0.1545	7164	0.0123	0.0295	−0.4333	0.3851
Change in interest rate spread vs. US (%)	11472	−0.0289	2.5691	−59.615	102.629	6141	−0.0108	0.7256	−17.909	13.096	5331	−0.0497	3.6876	−59.615	102.629
Inflation	14311	0.1226	0.3277	−0.0998	3.4593	6541	0.0322	0.0310	−0.0159	0.2714	7770	0.1987	0.4293	−0.0998	3.4593
Public debt to GDP	11751	0.5331	0.3222	0.0000	2.1860	6412	0.6316	0.2963	0.0981	2.1860	5339	0.4149	0.3122	0.0000	1.8394

Table 4 (*continued*)

Variable	Full sample *Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*	Advanced economies *Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*	Emerging economies *Obs*	*Mean*	*Std. Dev.*	*Min*	*Max*
Net foreign assets to GDP	9708	−0.1391	0.4582	−2.6882	2.8761	5458	−0.1194	0.4328	−2.6882	1.4620	4250	−0.1644	0.4877	−1.0504	2.8761
Current account to GDP	14323	−0.0024	0.0601	−0.4061	0.2542	6481	−0.0012	0.0521	−0.4061	0.1948	7842	−0.0034	0.0660	−0.2546	0.2542
Foreign exchange reserves to imports	13343	0.3213	0.2513	−0.1350	1.8182	6181	0.2213	0.1991	−0.1350	1.5271	7162	0.4075	0.2596	0.0223	1.8182
Country risk rating	13117	75.536	10.489	40.583	94.000	5725	83.098	5.119	56.500	94.000	7392	69.679	9.816	40.583	92.400
Banking crisis	13728	0.0193	0.1121	0.0000	1.0000	6072	0.0100	0.0817	0.0000	1.0000	7656	0.0266	0.1308	0.0000	1.0000
Rule of law	7540	0.8237	0.8710	−1.5900	2.1200	3335	1.5853	0.3567	0.3400	2.1200	4205	0.2196	0.6591	−1.5900	1.7900
Bid–ask spread (basis points)	11085	12.327	20.2244	−0.3602	494.31	5873	6.935	5.5511	−0.0003	57.46	5212	18.403	27.6696	−0.3602	494.31
Weight in world GDP at PPP	14609	2.0878	4.1261	0.0156	23.695	6624	3.2799	5.6571	0.0156	23.695	7985	1.0989	1.5635	0.0284	12.725
Stock market capit. to GDP	11968	0.6446	0.6856	0.0002	7.4250	5397	0.7289	0.5381	0.0066	3.4029	6571	0.5755	0.7797	0.0002	7.4250
Private credit to GDP	12811	0.7523	0.4751	0.0643	2.6976	6270	0.9950	0.4221	0.2585	2.6976	6541	0.5196	0.4009	0.0643	1.7676
Stock market capit. to world GDP	11968	1.4636	4.0517	0.0001	38.364	5397	2.5626	5.7411	0.0015	38.364	6571	0.5610	1.0121	0.0001	14.966
Private credit to world GDP	12811	1.9726	5.5716	0.0027	42.859	6270	3.6173	7.6095	0.0060	42.859	6541	0.3960	0.4700	0.0027	2.690
Financial openness	9708	2.4324	2.9107	0.1070	25.907	5458	3.0426	3.0906	0.5008	25.907	4250	1.6488	2.4487	0.1070	23.903
International debt to GDP	12269	0.2273	0.3167	0.0002	3.4439	5723	0.3726	0.4039	0.0235	3.4439	6546	0.1002	0.1037	0.0002	0.8895
Foreign loans to GDP	4758	0.0157	0.0305	0.0000	0.4021	2634	0.0236	0.0388	0.0000	0.4021	2124	0.0059	0.0070	0.0000	0.0593
Capital account restrictions	5566	0.2785	0.3209	0.0000	1.0000	2783	0.0714	0.1154	0.0000	0.7083	2783	0.4856	0.3268	0.0000	1.0000

Note: See Table 3 for a description of the variables. The sample period goes from January 1986 to December 2009 or longest available.

Table 5
Baseline estimation.

	(1)	(2)	(3)	(4) Advanced	(5) Emerging	(6) Until Aug-2007	(7) From Aug-2007	(8) Before euro	(9) After euro
Lagged dependent variable	0.329***	0.333***	0.331***	0.332***	0.323***	0.324***	0.254***	0.317***	0.322***
	(0.038)	(0.038)	(0.036)	(0.035)	(0.050)	(0.040)	(0.076)	(0.059)	(0.041)
Int. rate spread vs. US (lag)*GRA shock	−0.001	−0.000	0.000	−0.032	0.005	−0.001	−0.085***	−0.004	−0.012
	(0.004)	(0.004)	(0.005)	(0.028)	(0.004)	(0.004)	(0.022)	(0.006)	(0.012)
Int. rate spread vs. US (lag)	0.005	0.005	0.003	0.053	−0.001	0.002	−0.052	0.024	−0.007
	(0.011)	(0.011)	(0.010)	(0.045)	(0.010)	(0.011)	(0.060)	(0.015)	(0.024)
Z_t*GRA shock	1.511*	1.542*	1.770**	1.059	2.261***	0.536	2.933***	−0.837	3.412***
	(0.779)	(0.793)	(0.738)	(0.809)	(0.684)	(0.509)	(0.648)	(0.529)	(0.596)
Z_t	0.700	0.679	0.140	0.080	0.226	0.450	−9.802**	0.678	−2.277
	(0.710)	(0.694)	(0.732)	(0.926)	(0.713)	(0.510)	(3.808)	(0.593)	(1.511)
Growth of FX reserves*GRA shock	0.006								
	(2.338)								
Growth of FX reserves	7.997***								
	(1.907)								
Growth of FX reserves*Peg to the USD*GRA shock		1.935							
		(2.588)							
Growth of FX reserves*Peg to the USD		9.338***							
		(3.478)							
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.056**	0.056**	0.060***	0.267***	0.053**	0.051**	0.115*	0.059**	0.014
	(0.022)	(0.023)	(0.022)	(0.093)	(0.022)	(0.022)	(0.063)	(0.028)	(0.036)
Δ(Int. rate spread vs. USD)$_t$	−0.039	−0.038	−0.041	0.204***	−0.051	−0.044	−0.102	−0.055	−0.003
	(0.037)	(0.037)	(0.036)	(0.070)	(0.040)	(0.037)	(0.164)	(0.045)	(0.066)
Pegged to the USD*GRA shock	0.246	0.236	0.315*	0.158	0.402**	0.013	0.633***	−0.410**	0.511***
	(0.169)	(0.162)	(0.174)	(0.212)	(0.161)	(0.107)	(0.176)	(0.190)	(0.142)
Pegged to the USD	0.065	−0.026	0.015	−0.070	0.051	−0.014	0.855***	0.238	0.002
	(0.115)	(0.131)	(0.114)	(0.168)	(0.130)	(0.124)	(0.295)	(0.336)	(0.153)
Pegged to the EUR*GRA shock	−0.019	−0.020							
	(0.129)	(0.130)							
Pegged to the EUR	0.233	0.219							
	(0.167)	(0.169)							
Pegged to the DEM*GRA shock	0.526***	0.528***	0.541**	0.487**	0.553**	0.473**		0.335*	
	(0.178)	(0.180)	(0.203)	(0.195)	(0.223)	(0.184)		(0.167)	
Pegged to the DEM	0.099	0.107	0.014	0.020	−0.192	0.024		0.044	
	(0.164)	(0.164)	(0.172)	(0.167)	(0.213)	(0.169)		(0.162)	
Standardized values of GRA shock	−0.403*	−0.410*	−0.471**	−0.217	−0.574***	−0.138	−0.625**	0.307	−0.624***
	(0.213)	(0.205)	(0.209)	(0.183)	(0.198)	(0.114)	(0.281)	(0.191)	(0.196)
Observations	9083	9083	9580	4379	5201	8501	1079	4564	5016
Number of countries	51	51	51	22	29	51	39	43	41
R2 Within	0.144	0.142	0.145	0.150	0.151	0.120	0.375	0.135	0.192

Notes: The dependent variable is the currency excess return against the US dollar. 'GRA shock' indicates the global reversion shock that leads to an increase in the VIX and a fall in the return of a global unhedged stock market portfolio. The variable Z_t is the recursive correlation between currency excess returns and global risk aversion shocks up to time $t-1$. The panel is estimated through OLS fixed effects. Driscoll and Kraay (1998) robust standard errors accounting for temporal and cross-sectional dependence are reported in parentheses. ***, **, * indicate statistical significance at the 1, 5, 10% level, respectively. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability.

Table 6
Including measures of country risk and vulnerability.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Lagged dependent variable	0.330***	0.340***	0.340***	0.330***	0.329***	0.330***	0.333***	0.312***	0.345***
	(0.036)	(0.031)	(0.032)	(0.036)	(0.037)	(0.038)	(0.039)	(0.048)	(0.032)
Int. rate spread vs. US (lag)*GRA shock	−0.003	−0.015	−0.003	0.003	0.000	0.005	−0.001	−0.003	−0.000
	(0.009)	(0.010)	(0.006)	(0.005)	(0.004)	(0.004)	(0.004)	(0.010)	(0.006)
Int. rate spread vs. US (lag)	0.000	−0.000	0.001	0.003	0.005	0.001	0.003	0.001	0.000
	(0.014)	(0.018)	(0.016)	(0.010)	(0.011)	(0.011)	(0.011)	(0.019)	(0.018)
Z_t*GRA shock	1.771**	1.948***	2.236***	1.807**	1.614*	1.369*	1.497**	3.286***	1.901**
	(0.729)	(0.701)	(0.745)	(0.760)	(0.838)	(0.745)	(0.739)	(0.677)	(0.827)
Z_t	0.026	−0.286	−0.530	0.093	0.777	0.606	0.675	−1.486	−0.009
	(0.740)	(0.832)	(0.931)	(0.753)	(0.659)	(0.584)	(0.612)	(1.657)	(0.808)
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.054**	0.080*	0.038*	0.063***	0.058**	0.064***	0.059***	0.035	0.039*
	(0.021)	(0.044)	(0.020)	(0.023)	(0.022)	(0.022)	(0.022)	(0.028)	(0.020)
Δ(Int. rate spread vs. USD)$_t$	−0.042	−0.059	0.007	−0.041	−0.040	−0.041	−0.040	−0.042	0.009
	(0.036)	(0.074)	(0.028)	(0.036)	(0.037)	(0.036)	(0.036)	(0.066)	(0.028)
Pegged to the USD*GRA shock	0.314*	0.322**	0.262*	0.235	0.236	0.264	0.264	0.448***	0.204*
	(0.167)	(0.159)	(0.135)	(0.142)	(0.158)	(0.170)	(0.179)	(0.137)	(0.116)
Pegged to the USD	0.015	−0.006	0.012	0.023	0.080	0.065	0.049	0.013	0.037
	(0.113)	(0.103)	(0.114)	(0.118)	(0.122)	(0.119)	(0.117)	(0.137)	(0.119)
Pegged to the DEM*GRA shock	0.541**	0.470**	0.445**	0.517**	0.536**	0.480**	0.543**	0.877**	0.422*
	(0.204)	(0.202)	(0.210)	(0.198)	(0.204)	(0.201)	(0.207)	(0.357)	(0.212)
Pegged to the DEM	0.000	0.015	−0.040	0.033	0.038	0.029	0.037	−0.268	−0.018
	(0.172)	(0.176)	(0.181)	(0.172)	(0.173)	(0.174)	(0.174)	(0.303)	(0.179)
Standardized values of GRA shock	−0.486**	−0.549**	−0.474**	−0.464**	−0.463	−1.307***	−0.418*	°0.592***	−0.703**
	(0.224)	(0.244)	(0.211)	(0.206)	(0.287)	(0.221)	(0.217)	(0.174)	(0.323)
Inflation (lag12)* GRA shock	0.497								
	(0.949)								
Inflation (lag12)	0.534								
	(0.907)								
Public debt to GDP (lag12)*GRA shock		0.226							
		(0.135)							
Public debt to GDP (lag12)		−0.002							
		(0.250)							
Net foreign assets to GDP (lag12)*GRA shock			0.297***						0.238***
			(0.074)						(0.051)
Net foreign assets to GDP (lag12)			0.195						0.035
			(0.240)						(0.347)
Curr. account to GDP (lag12)*GRA shock				1.708**					0.233
				(0.662)					(0.490)
Curr. account to GDP (lag12)				1.338					0.624
				(1.016)					(1.449)
FX reserves to imports (lag)*GRA shock					0.121				
					(0.198)				
FX reserves to imports (lag)					0.436				
					(0.265)				
Country rating (lag12)*GRA shock						0.012***			0.004
						(0.004)			(0.006)
Country rating (lag12)						−0.021			−0.018
						(0.015)			(0.020)
Banking crisis (lag12)*GRA shock							0.522		
							(0.416)		
Banking crisis (lag12)							0.038		
							(0.870)		
Rule of law (lag12)*GRA shock								−0.015	
								(0.061)	
Rule of law (lag12)								−0.805	
								(0.604)	
Observations	9550	8562	7251	9562	9121	8909	9164	5561	6602
Number of countries	51	50	49	51	51	49	51	51	47
R2 Within	0.145	0.163	0.169	0.147	0.139	0.140	0.141	0.159	0.161

See notes to Tables 5 and 3 for a description of the variables and the estimation method. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability.

Table 7
Including measures of size and liquidity of financial and foreign exchange markets.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Lagged dependent variable	0.334***	0.340***	0.334***	0.328***	0.337***	0.330***	0.321***
	(0.031)	(0.032)	(0.035)	(0.029)	(0.035)	(0.029)	(0.031)
Int. rate spread vs. US (lag)*GRA shock	−0.003	−0.002	−0.006	−0.004	−0.002	−0.000	−0.004
	(0.006)	(0.006)	(0.006)	(0.008)	(0.006)	(0.007)	(0.008)
Int. rate spread vs. US (lag)	−0.005	0.001	−0.001	−0.008	−0.001	−0.008	−0.011
	(0.018)	(0.016)	(0.017)	(0.020)	(0.017)	(0.020)	(0.021)
Z_t*GRA shock	2.433***	2.233***	3.100***	2.413***	3.216***	2.371***	3.312***

Table 7 (*continued*)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	(0.712)	(0.750)	(0.561)	(0.848)	(0.634)	(0.800)	(0.614)
Z_t	−0.693	−0.568	−2.440**	−0.467	−2.107*	−0.479	−2.731**
	(0.951)	(0.935)	(1.115)	(0.934)	(1.166)	(0.970)	(1.267)
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.022	0.039*	0.024	0.012	0.032	0.018	0.005
	(0.020)	(0.020)	(0.019)	(0.022)	(0.019)	(0.022)	(0.021)
Δ(Int. rate spread vs. USD)$_t$	0.027	0.007	0.006	0.041	0.007	0.043	0.040
	(0.032)	(0.028)	(0.029)	(0.036)	(0.029)	(0.035)	(0.036)
Pegged to the USD*GRA shock	0.264**	0.262*	0.252**	0.205*	0.324***	0.260**	0.253**
	(0.128)	(0.132)	(0.103)	(0.109)	(0.113)	(0.125)	(0.100)
Pegged to the USD	0.038	0.017	0.008	−0.016	0.044	−0.017	−0.005
	(0.126)	(0.116)	(0.109)	(0.120)	(0.115)	(0.125)	(0.124)
Pegged to the DEM*GRA shock	0.442*	0.453**	0.245	0.386*	0.440	0.390*	0.201
	(0.223)	(0.210)	(0.269)	(0.205)	(0.272)	(0.208)	(0.279)
Pegged to the DEM	−0.071	−0.044	−0.060	−0.054	−0.006	−0.057	−0.078
	(0.190)	(0.180)	(0.223)	(0.177)	(0.236)	(0.182)	(0.220)
Standardized values of GRA shock	−0.477**	−0.510**	−0.221	−0.330**	−0.578***	−0.493**	−0.181
	(0.230)	(0.217)	(0.141)	(0.163)	(0.197)	(0.202)	(0.135)
Net foreign assets to GDP (lag12)*GRA shock	0.281***	0.287***	0.556***	0.329***	0.237***	0.268***	0.600**
	(0.079)	(0.073)	(0.194)	(0.083)	(0.079)	(0.067)	(0.225)
Net foreign assets to GDP (lag12)	0.280	0.206	0.206	0.208	0.205	0.174	0.161
	(0.252)	(0.243)	(0.284)	(0.261)	(0.276)	(0.297)	(0.383)
Bid–ask spread (lag)*GRA shock	−0.002						
	(0.004)						
Bid–ask spread (lag)	0.001						
	(0.005)						
Weight in world GDP (lag12)*GRA shock		0.018***					−0.061
		(0.006)					(0.037)
Weight in world GDP (lag12)		0.085					0.256
		(0.102)					(0.184)
Stock mkt. capital. to GDP (lag12)*GRA shock			−0.310***				−0.368**
			(0.115)				(0.137)
Stock mkt. capital. to GDP (lag12)			−0.103				−0.162
			(0.193)				(0.213)
Private credit to GDP (lag12)*GRA shock				−0.150			
				(0.109)			
Private credit to GDP (lag12)				0.045			
				(0.215)			
Stock mkt capital. to world GDP (lag12)*GRA shock					0.014*		0.067**
					(0.007)		(0.032)
Stock mkt capital. to world GDP (lag12)					0.029		0.045
					(0.049)		(0.050)
Private credit to world GDP (lag12)*GRA shock						0.012*	0.015
						(0.007)	(0.028)
Private credit to world GDP (lag12)						−0.024	−0.071
						(0.040)	(0.051)
Observations	6687	7251	6348	7060	6315	6994	6116
Number of countries	48	49	49	48	49	48	48
R2 Within	0.172	0.169	0.184	0.163	0.180	0.163	0.179

See notes to Table 5 and Table 3 for a description of the variables and the estimation method. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability.

Table 8
Including measures of financial openness.

	(1)	(2)	(3)	(4)	(5)	(6)
Lagged dependent variable	0.337***	0.329***	0.336***	0.353***	0.353***	0.326***
	(0.031)	(0.032)	(0.036)	(0.045)	(0.045)	(0.031)
Int. rate spread vs. US (lag)*GRA shock	−0.005	−0.001	−0.004	0.003	0.003	−0.003
	(0.006)	(0.006)	(0.007)	(0.005)	(0.005)	(0.006)
Int. rate spread vs. US (lag)	0.001	−0.006	−0.006	−0.004	−0.004	−0.006
	(0.016)	(0.019)	(0.021)	(0.021)	(0.021)	(0.019)
Z_t*GRA shock	2.280***	3.409***	4.014***	1.929***	1.935***	3.449***
	(0.828)	(0.663)	(0.774)	(0.615)	(0.617)	(0.689)
Z_t	−0.628	−2.454**	−1.898	−2.742	−2.718	−2.542**
	(0.920)	(1.149)	(1.539)	(2.434)	(2.410)	(1.101)
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.035*	0.015	0.039	0.024	0.023	0.013
	(0.020)	(0.020)	(0.035)	(0.023)	(0.023)	(0.020)
Δ(Int. rate spread vs. USD)$_t$	0.007	0.036	0.020	0.007	0.007	0.035

(continued on next page)

Table 8 (*continued*)

	(1)	(2)	(3)	(4)	(5)	(6)
	(0.028)	(0.035)	(0.050)	(0.029)	(0.029)	(0.036)
Pegged to the USD*GRA shock	0.220*	0.308***	0.266**	0.156	0.155	0.285***
	(0.119)	(0.113)	(0.113)	(0.098)	(0.098)	(0.106)
Pegged to the USD	0.009	0.035	0.008	−0.165	−0.166	0.029
	(0.112)	(0.124)	(0.163)	(0.178)	(0.178)	(0.123)
Pegged to the DEM*GRA shock	0.337*	0.430*	0.630**	0.758**	0.760**	0.359
	(0.197)	(0.228)	(0.294)	(0.288)	(0.287)	(0.219)
Pegged to the DEM	−0.082	0.099	−0.460	−0.329	−0.327	0.057
	(0.174)	(0.206)	(0.291)	(0.269)	(0.270)	(0.196)
Standardized values of GRA shock	−0.268*	−0.492**	−0.672***	−0.341**	−0.347**	−0.348**
	(0.146)	(0.186)	(0.231)	(0.146)	(0.146)	(0.134)
Net foreign assets to GDP (lag12)*GRA shock	0.521***	0.233***	0.240***	0.195***	0.195***	0.433**
	(0.174)	(0.070)	(0.083)	(0.072)	(0.072)	(0.167)
Net foreign assets to GDP (lag12)	0.176	0.291	0.313	0.091	0.099	0.329
	(0.221)	(0.223)	(0.383)	(0.288)	(0.304)	(0.272)
Weight in world GDP (lag12)*GRA shock	0.005	0.017***	0.043*	0.008	0.009	0.007
	(0.006)	(0.006)	(0.022)	(0.008)	(0.008)	(0.006)
Weight in world GDP (lag12)	0.073	0.142	−0.224	0.047	0.048	0.144
	(0.097)	(0.109)	(0.210)	(0.278)	(0.277)	(0.110)
Financial openness (lag12)*GRA shock	−0.063**					−0.049*
	(0.026)					(0.025)
Financial openness (lag12)	−0.019					−0.044
	(0.052)					(0.075)
International debt to GDP (lag12)*GRA shock		−0.389**				−0.172
		(0.164)				(0.128)
International debt to GDP (lag12)		−0.158				0.050
		(0.266)				(0.368)
Foreign loans to GDP (lag12)*GRA shock			−2.976**			
			(1.253)			
Foreign loans to GDP (lag12)			−0.822			
			(3.008)			
Capital restrictions (lag12)*GRA shock				−0.057		
				(0.128)		
Capital restrictions (lag12)				0.073		
				(0.416)		
Inflow restrictions (lag12)*GRA shock					−0.043	
					(0.141)	
Inflow restrictions (lag12)					0.078	
					(0.445)	
Observations	7251	6619	2541	3242	3242	6619
Number of countries	49	49	47	44	44	49
R2 Within	0.172	0.182	0.253	0.143	0.143	0.183

See notes to Tables 5 and 3 for a description of the variables and the estimation method. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability.

Table 9
Robustness of the final equation — benchmarking, asymmetry and linearity.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Baseline	Random walk	Carry trade	Fundamentals	No crisis	Crisis	Neg. GRA shock	Pos. GRA shock	Very high GRA shock
Lagged dependent variable	0.337***	0.361***	0.353***	0.350***	0.333***	0.262***	0.316***	0.357***	0.502***
	(0.031)	(0.039)	(0.034)	(0.031)	(0.036)	(0.065)	(0.060)	(0.060)	(0.102)
Int. rate spread vs. US (lag)*GRA shock	−0.005		−0.027**		−0.015*	0.001	−0.014	0.004	0.016
	(0.006)		(0.011)		(0.009)	(0.009)	(0.018)	(0.009)	(0.022)
Int. rate spread vs. US (lag)	0.001		−0.011		−0.009	0.054**	−0.007	−0.010	−0.026
	(0.016)		(0.019)		(0.019)	(0.023)	(0.022)	(0.019)	(0.053)
Z_t*GRA shock	2.282***				1.201**	3.298***	0.370	3.057**	1.775**
	(0.826)				(0.478)	(0.903)	(1.487)	(1.206)	(0.881)
Z_t	−0.606				0.086	−4.532*	−0.932	−3.192**	−0.268
	(0.913)				(0.777)	(2.374)	(1.243)	(1.356)	(5.390)
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.035*				0.027	0.123***	−0.039	0.005	0.105
	(0.020)				(0.036)	(0.036)	(0.112)	(0.028)	(0.063)
Δ(Int. rate spread vs. USD)$_t$	0.007				0.007	−0.202**	−0.058	0.057	−0.148
	(0.028)				(0.029)	(0.083)	(0.089)	(0.043)	(0.142)
Pegged to the USD*GRA shock	0.219*				−0.075	0.441***	0.178	0.260**	0.186
	(0.120)				(0.124)	(0.138)	(0.232)	(0.104)	(0.136)
Pegged to the USD	0.005				−0.046	0.451	−0.036	−0.034	0.198
	(0.112)				(0.104)	(0.367)	(0.153)	(0.154)	(0.329)
Pegged to the DEM*GRA shock	0.332*				0.421	0.130	0.651	−0.071	−0.158

Table 9 (*continued*)

	(1) Baseline	(2) Random walk	(3) Carry trade	(4) Fundamentals	(5) No crisis	(6) Crisis	(7) Neg. GRA shock	(8) Pos. GRA shock	(9) Very high GRA shock
	(0.196)				(0.266)	(0.361)	(0.528)	(0.271)	(0.129)
Pegged to the DEM	−0.083				−0.085	0.435	0.052	0.184	1.242**
	(0.173)				(0.195)	(0.723)	(0.366)	(0.288)	(0.490)
Standardized values of GRA shock	−0.253*		−0.163	0.011	−0.080	−0.688***	−0.838**	−0.365	−0.658***
	(0.147)		(0.171)	(0.111)	(0.169)	(0.228)	(0.372)	(0.245)	(0.201)
NFA to GDP (lag12)*GRA shock	0.529***			0.728***	0.041	0.755***	−0.475	0.970***	1.328***
	(0.168)			(0.197)	(0.132)	(0.222)	(0.421)	(0.120)	(0.222)
NFA to GDP (lag12)	0.165			0.184	0.061	0.727	−0.103	−0.944**	−1.651***
	(0.221)			(0.208)	(0.254)	(0.513)	(0.285)	(0.383)	(0.475)
Financial openness (lag12)*GRA shock	−0.064**			−0.076***	0.024	−0.080***	0.162*	−0.105***	−0.124***
	(0.025)			(0.026)	(0.025)	(0.027)	(0.081)	(0.021)	(0.039)
Financial openness (lag12)	−0.021			0.002	0.030	−0.197**	0.164***	−0.020	0.095
	(0.053)			(0.057)	(0.055)	(0.082)	(0.056)	(0.085)	(0.112)
Observations	7251	7550	7550	7550	6325	926	4177	3074	820
Number of countries	49	50	50	50	49	47	49	49	49
R2 Within	0.172	0.119	0.136	0.141	0.120	0.412	0.118	0.234	0.465

See notes to Tables 5 and 3 for a description of the variables and the estimation method. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability. Column (6) includes only distress episodes (Crisis). These begin in the month in which the percentage change in the VIX is above two standard deviations and end when the level of the VIX falls below its long-term average plus one standard deviation. Columns (7) and (8) split the sample into negative and positive Global Risk Aversion (GRA) shocks, respectively. Column (9) includes only observations when the global risk aversion shock is in the highest decile.

Table 10
Robustness of the final equation — time and country sample.

	(1) Baseline	(2) Advanced	(3) Emerging	(4) Until Aug-2007	(5) From Aug-2007	(6) Before euro	(7) After euro
Lagged dependent variable	0.337***	0.331***	0.332***	0.338***	0.252***	0.356***	0.314***
	(0.031)	(0.044)	(0.043)	(0.034)	(0.071)	(0.043)	(0.040)
Int. rate spread vs. US (lag)*GRA shock	−0.005	0.003	0.000	−0.002	−0.064***	−0.010*	−0.002
	(0.006)	(0.005)	(0.006)	(0.005)	(0.021)	(0.006)	(0.011)
Int. rate spread vs. US (lag)	0.001	−0.006	−0.003	−0.001	−0.047	0.043	−0.011
	(0.016)	(0.018)	(0.018)	(0.018)	(0.059)	(0.028)	(0.025)
Z_t*GRA shock	2.282***	3.203***	3.491***	0.672	3.373***	−0.795	3.661***
	(0.826)	(0.855)	(0.895)	(0.489)	(0.470)	(0.641)	(0.660)
Z_t	−0.606	−2.584*	−2.389*	−0.070	−9.407**	0.145	−3.538*
	(0.913)	(1.383)	(1.376)	(0.753)	(3.799)	(0.977)	(1.913)
Δ(Int. rate spread vs. USD)$_t$*GRA shock	0.035*	0.029	0.029	0.028	0.112*	0.045	0.008
	(0.020)	(0.021)	(0.020)	(0.021)	(0.062)	(0.028)	(0.035)
Δ(Int. rate spread vs. USD)$_t$	0.007	0.000	0.000	0.007	−0.088	0.016	0.026
	(0.028)	(0.030)	(0.030)	(0.028)	(0.166)	(0.029)	(0.054)
Pegged to the USD*GRA shock	0.219*	0.235**	0.262***	0.014	0.409**	−0.262	0.317***
	(0.120)	(0.087)	(0.085)	(0.087)	(0.155)	(0.167)	(0.111)
Pegged to the USD	0.005	0.116	0.097	−0.096	0.950**	−0.121	0.094
	(0.112)	(0.129)	(0.130)	(0.106)	(0.357)	(0.181)	(0.160)
Pegged to the DEM*GRA shock	0.332*	0.289	0.283	0.421**		0.222	
	(0.196)	(0.233)	(0.234)	(0.184)		(0.154)	
Pegged to the DEM	−0.083	−0.181	−0.198	0.003		−0.020	
	(0.173)	(0.265)	(0.269)	(0.169)		(0.167)	
Standardized values of GRA shock	−0.253*	−0.369***	−0.375***	−0.134	−0.407	0.381	−0.345**
	(0.147)	(0.095)	(0.093)	(0.114)	(0.273)	(0.238)	(0.158)
NFA to GDP (lag12)*GRA shock	0.529***	0.496	0.533	0.187	0.430**	0.313*	0.534***
	(0.168)	(0.328)	(0.335)	(0.130)	(0.189)	(0.170)	(0.188)
NFA to GDP (lag12)	0.165	0.040	0.009	−0.123	0.804	−0.035	0.461*
	(0.221)	(0.526)	(0.524)	(0.290)	(0.836)	(0.621)	(0.256)
Financial openness (lag12)*GRA shock	−0.064**	−0.052	−0.058	0.007	−0.054***	0.010	−0.073***
	(0.025)	(0.037)	(0.037)	(0.026)	(0.018)	(0.047)	(0.025)
Financial openness (lag12)	−0.021	−0.008	0.000	0.044	−0.264	−0.238	−0.034
	(0.053)	(0.066)	(0.066)	(0.072)	(0.226)	(0.266)	(0.067)
Observations	7251	3420	3506	6211	1040	2839	4412
Number of countries	49	27	28	49	38	33	40
R2 Within	0.172	0.186	0.195	0.125	0.383	0.158	0.205

See notes to Tables 5 and 3 for a description of the variables and the estimation method. The sample period is January 1986 to December 2009, unless otherwise stated and depending on data availability.

Table 11
Safe haven performance of selected portfolio returns.

Returns	*Performance during crisis periods*
Global unhedged stock market	−0.04*** (0.001)
Excess returns on the euro vs. USD	−0.00 (0.01)
Excess returns on the Japanese yen vs. USD	0.01*** (0.00)
Excess returns on the Swiss franc vs. USD	0.008* (0.005)
High-NFA portfolio (a)	0.02** (0.01)
Low interest rate portfolio (b)	0.05*** (0.02)
(Memo: VIX)	7.74*** (0.808)

Note: The table reports the coefficient *a2* in an estimated equation,

$$r = c + a1 * r(-1) + a2 * Crisis + a3 * Dum99$$

where r is the selected return (e.g. global unhedged stock market), *Crisis* is a dummy variable indicating the presence of a crisis (same as in Table 9, column (Christiansen et al., 2011)), and Dum99 is a dummy variable taking values 1 after 1999:1 and zero otherwise, included to capture the potential effect of the change of composition of the panel with the introduction of the euro in January 1999. */**/*** indicates significance at the 10%/5%/1% confidence level respectively. Sample period: 1986:1–2009:12. Note that the reported coefficients are not in percentage points. Excess returns are defined as in text, equation (x).

a) Portofolio computed assuming, at time t, equal-weighted *long* position on currencies with NFA position at least one standard deviation *above* the average across all currencies at time $t-12$, and an equal-weighted *short* position on currencies with NFA position at least one standard deviation *below* the average across all currencies at time $t-12$.

b) Portofolio computed assuming, at time t, equal-weighted *long* position on currencies with an interest rate spread vs. the US at least one standard deviation *below* the average across all currencies at time $t-12$, and an equal-weighted *short* position on currencies with an interest rate spread vs. the US at least one standard deviation *above* the average across all currencies at time $t-12$.

References

Bansal, R., Dahlquist, M., 2000. The forward premium puzzle: different tales from developed and emerging economies. Journal of International Economics 51 (1), 115–144.

Brunnermeier, M.K., Nagel, S., Pedersen, L.H., 2008. Carry trade and currency crashes. NBER Working Paper No. 14473, November.

Campbell, J.Y., Serfaty-de Medeiros, K., Viceira, L.M., 2010. Global currency hedging. Journal of Finance LXV 1, 87–121.

Cheung, Y.-W., Chinn, M.D., Pascual, A.G., 2005. Empirical exchange rate models of the nineties: are any fit to survive? Journal of International Money and Finance 24, 1150–1175.

Chinn, M.D., 2006. The (partial) rehabilitation of interest parity in the floating era: longer horizons, alternative expectations, and emerging markets. Journal of International Money and Finance 25, 7–21.

Christiansen, C., Ranaldo, A., Soderlind, P., 2011. The time-varying risk of carry trade strategies. Journal of Financial and Quantitative Analysis 46 (4), 1107–1125.

Collin-Dufresne, P., Goldstein, R.S., Martin, J.S., 2001. The determinants of credit spread changes. Journal of Finance 56 (6), 2177–2207.

Diekmann, B., Meurers, M., 2007. The US dollar: safe haven. Despite rising US current account deficit. Working Paper Series — Deutsche Bank Research Notes 25, August.

Driscoll, J.C., Kraay, A.C., 1998. Consistent covariance matrix estimation with spatially dependent panel data. The Review of Economics and Statistics 80, 549–560.

Engel, C., West, K.D., 2005. Exchange rates and fundamentals. Journal of Political Economy 113 (3), 485–517.

Fratzscher, M., 2009. What explains global exchange rate movements during the financial crisis? ECB Working Paper No. 1060, June.

Hui, C., Genberg, H., Chung, T., 2010. Funding liquidity risk and deviations from interest-rate parity during the financial crisis of 2007–2009. International Journal of Finance & Economics 16 (4), 307–323.

Jorda, O., Taylor, A.M., 2009. The carry trade and fundamentals: nothing to fear but FEER itself. NBER Working Paper No. 15518, November.

Kaul, A., Sapp, S., 2006. Y2K fears and safe haven trading of the U.S. dollar. Journal of International Money and Finance 25 (5), 760–779.

Kohler, M., 2010. Exchange rates during financial crises. BIS Quarterly Review, March.

Laeven, L.A., Valencia, F.V., 2008. Systemic banking crises: a new database. IMF Working Papers 224/08.

Levy-Yeyati, E., Sturzenegger, F., 2005. Classifying exchange rate regimes: deeds vs words. European Economic Review 49, 1603–1635.

Lustig, H., Roussanov, N., Verdelhan, A., 2011. Common risk factors in currency markets. Review of Financial Studies 24 (11), 3731–3777.

McCauley, R.N., McGuire, P., 2009. Dollar appreciation in 2008: safe haven, carry trades, dollar shortage and overhedging. BIS Quarterly Review, December.

Meese, R.A., Rogoff, K.S., 1983. Empirical exchange rate models of the seventies: do they fit out of sample? Journal of International Economics 14, 3–24.

Menkhoff, L., Sarno, L., Schmeling, M., Schrimpf, A., 2012. Carry trades and global foreign exchange volatility. Journal of Finance 67 (2), 681–718.

Nickell, S., 1981. Biases in dynamic models with fixed effects. Econometrica 49, 1417–1426.

Nozaki, M., 2010. Do currency fundamentals matter for currency speculators? IMF Working Paper WP/10/39, February.

Ranaldo, A., Soderlind, P., 2009. Safe haven currencies. Review of Finance 14 (3), 385–407.

Rubio-Ramirez, J.F., Waggoner, D.F., Zha, T., 2010. Structural vector autoregressions: theory or identification and algorithms for inference. Review of Economic Studies 77 (2), 665–696.

Schindler, M., 2009. Measuring financial integration: a new data set. IMF Staff Papers 56 (1), 222–238.